

Part I

SUMMARY of Understanding Functions

Summary

February 21, 2026

What we need to know about functions

- State the definition of a function.
- Find the domain and range of a function.
- Distinguish between functions by considering their domains.
- Recognize different representations of the same function.
- Determine where a function is positive or negative.
- Determine algebraically whether a function is even, odd, or neither.
- Use symmetry when graphing even or odd functions.
- Plot basic functions.
- Apply appropriate transformations to graphs of basic functions (vertical and horizontal shifts, vertical and horizontal stretching and reflecting)
- Perform basic operations and compositions on functions.
- Work with piecewise defined functions.
- Determine if a function is one-to-one.
- Define and work with inverse functions.
- Plot inverses of basic functions.
- Find inverse functions (algebraically and graphically).
- Find the largest interval containing a given point where the function is invertible.
- Determine the intervals on which a function has an inverse.
- Relate the domain and range of f and the domain and range of f^{-1} .
- Use a sign-chart to determine where the function is positive/negative.

Part II

Recitation Questions

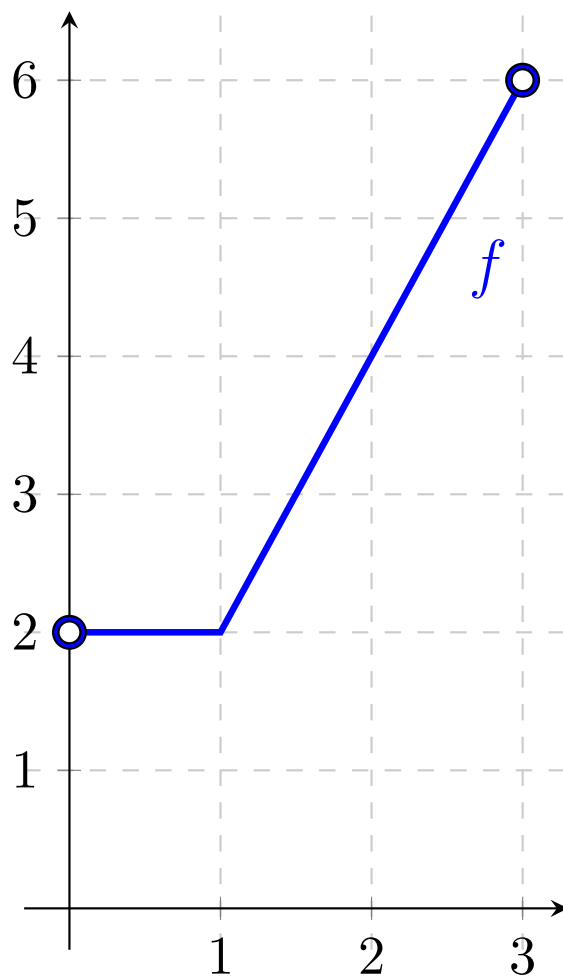
Problem 1

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Problem 1

Problem 1 Find a formula for the function f whose graph is given in the figure below.

What is the domain of f ? What is the range of f ? Is the function f linear?



Explanation. $f(x) = \begin{cases} 2 & \text{if } 0 < x \leq 1 \\ 2 + 2(x - 1) & \text{if } 1 < x < 3 \end{cases}$

Domain of $f = (0, 3)$, range of $f = [2, 6)$.

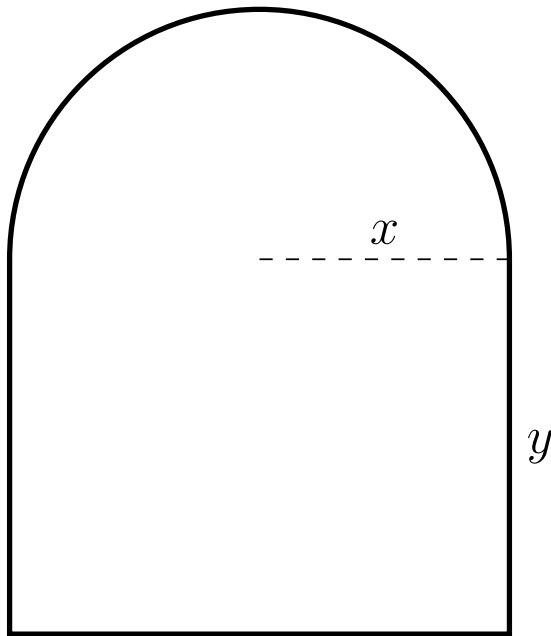
f is not linear; because its graph is NOT a line; it is a piecewise defined function.

Problem 2

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Problem 2

Problem 2 A mirror has the shape of a rectangle surmounted by a semicircle (see figure). The area of the mirror is 32 in^2 . Let x be the radius of the semicircle (that lies on top of the rectangle). Express the perimeter of the mirror P (in inches) as a function of x (in inches). Is P a polynomial, a rational function or a transcendental function?



Explanation. Since the area $A = 32 = 2xy + \frac{x^2\pi}{2}$, it follows that $y = \frac{16}{x} - \frac{x\pi}{4}$ and that $P(x) = 2x + 2y + x\pi = 2x + \frac{32}{x} - \frac{x\pi}{2} + x\pi = 2x + \frac{x\pi}{2} + \frac{32}{x}$

Problem 3

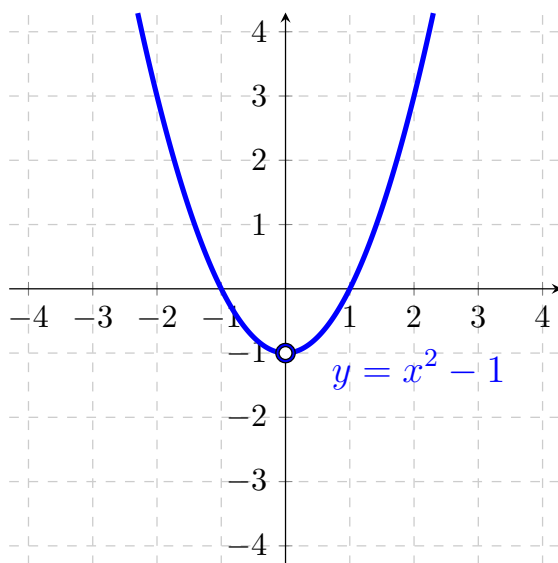
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Problem 3 Define $f(x) = \begin{cases} x^2 - 1 & \text{if } x < 0 \\ ? & \text{if } x > 0 \end{cases}$

(a) Find an expression for "?" such that f will be even.

Explanation. If f is even then $f(x) = f(-x)$.

$$\begin{aligned} f(x) &= f(-x) \text{ for } x > 0 \\ &= (-x)^2 - 1 \\ &= x^2 - 1 \end{aligned}$$

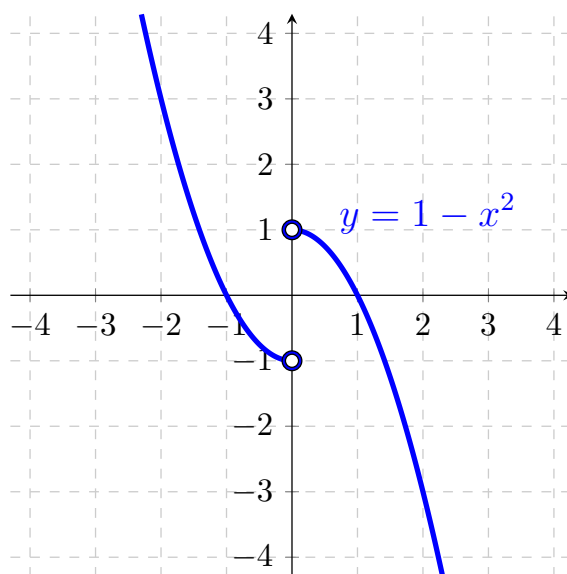


(b) Find an expression for "?" such that f will be odd.

Explanation. If f is odd then $f(-x) = -f(x)$.

$$\begin{aligned} -f(x) &= f(-x) \text{ for } x > 0 \\ &= (-x)^2 - 1 \\ &= x^2 - 1 \\ \implies f(x) &= -x^2 + 1 \end{aligned}$$

Problem 3



Problem 4

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Problem 4 Given $y(t) = t - \frac{\pi}{3}$ and $w(t) = \sin(t)$. Find:

(a) $y(w(t))$

Explanation. $y(w(t)) = y(\sin(t)) = \sin(t) - \frac{\pi}{3}$

(b) $w(y(t))$

Explanation. $w(y(t)) = w\left(t - \frac{\pi}{3}\right) = \sin\left(t - \frac{\pi}{3}\right)$

(c) $w\left(y\left(\frac{4\pi}{3}\right)\right)$

Explanation. $w\left(y\left(\frac{4\pi}{3}\right)\right) = \sin\left(\frac{4\pi}{3} - \frac{\pi}{3}\right) = \sin(\pi) = 0$

(d) $y(w(\frac{4\pi}{3}))$

Explanation. $y\left(w\left(\frac{4\pi}{3}\right)\right) = \sin\left(\frac{4\pi}{3}\right) - \frac{\pi}{3} = \frac{-\sqrt{3}}{2} - \frac{\pi}{3}$

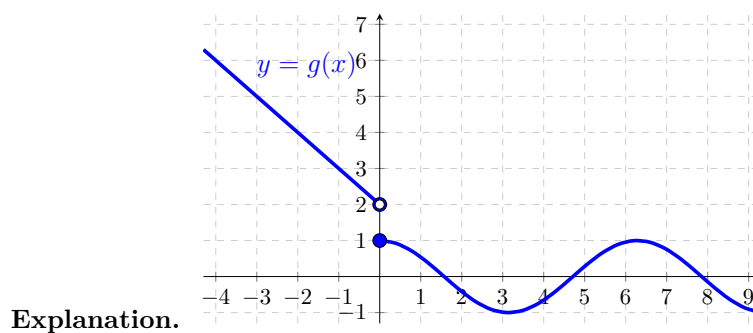
You should know values of $\sin(x)$ and $\cos(x)$ for all values found on the standard unit circle.

Problem 5

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Problem 5 Define $g(x) = \begin{cases} |x - 2| & \text{if } x < 0 \\ \cos(x) & \text{if } x \geq 0 \end{cases}$

(a) Sketch a graph of g



(b) Find the domain and range of g

Explanation. Domain: $(-\infty, \infty)$, Range: $[-1, 1] \cup (2, \infty)$

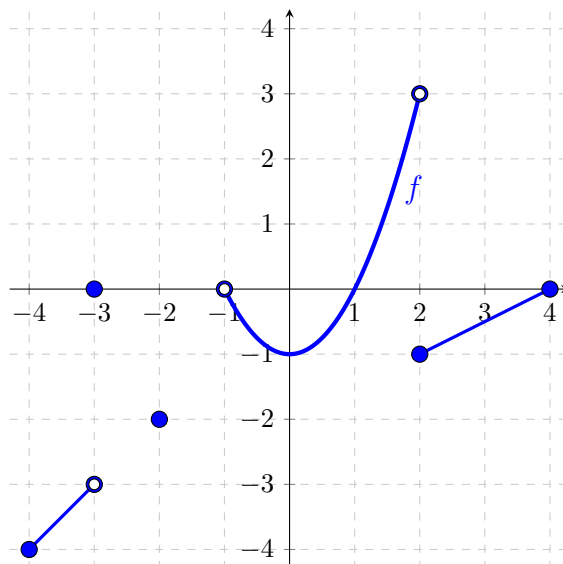
(c) Find the values of $g(\pi)$ and $g(-\pi)$

Explanation. $g(\pi) = \cos(\pi) = -1$ and $g(-\pi) = |-\pi - 2| = \pi + 2$

Problem 6

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Problem 6 The entire graph of $f(x)$ is given below.



- (a) Find the domain and range of f

Explanation. Domain: $[-4, -3] \cup \{-2\} \cup (-1, 4]$ Range: $[-4, -3] \cup \{-2\} \cup [-1, 3)$

- (b) Find the values of $f(-3)$, $f(-2)$, $f(-1)$, $f(2)$

Explanation. $f(-3) = 0$, $f(-2) = -2$, $f(-1)$ does not exist, $f(2) = -1$

- (c) Find the intervals on which $f(x)$ is positive. Find the intervals on which $f(x)$ is negative.

Explanation. $f(x)$ is positive on $(1, 2)$. $f(x)$ is negative on $[-4, -3)$, $\{-2\}$, $(-1, 1)$, $[2, 4)$

- (d) Find the intervals on which f is increasing. Find the intervals on which f is decreasing.

Explanation. f is increasing on $(-4, -3)$, $(0, 2)$, $(2, 4)$. f is decreasing on $(-1, 0)$. It would also be OK to say that f is increasing on $[-4, -3]$, $[0, 2)$, $[2, 4]$ and f is decreasing on $(-1, 0]$. Typically we will not be including endpoints of intervals when we talk about increasing/decreasing.

- (e) True or False: $f(1.5) < f(2)$

Explanation. False, $f(2) < f(1.5)$

Problem 7

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Problem 7 Determine if the function is even, odd, or neither.

(a) $h(x) = x^4 + x^2 - 3$

Explanation. A function is even if $f(x) = f(-x)$, for all x in the domain, which means its graph is symmetric about the y-axis. A function is odd if $f(-x) = -f(x)$, for all x in the domain, which means its graph is symmetric about the origin.

$$h(x) = x^4 + x^2 - 3$$

$$h(-x) = (-x)^4 + (-x)^2 - 3 = x^4 + x^2 - 3$$

$h(x) = h(-x)$. Hence h is even. This can be verified by graphing h and seeing that its graph is symmetric about the y-axis.

(b) $s(t) = t^2 - t$

Explanation.

$$s(t) = t^2 - t$$

$$s(-t) = (-t)^2 - (-t) = t^2 + t$$

This does not equal $s(t)$ so $s(t)$ is not even.

$$-s(t) = -(t^2 - t) = -t^2 + t$$

This does not equal $s(-t)$ so s is not odd. Hence, $s(t)$ is neither even, nor odd.

(c) We know that $\sin(\theta)$ is odd and $\cos(\theta)$ is even. Is $g(\theta) = \tan(\theta)$ even, odd, or neither?

Explanation.

$$g(\theta) = \tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$$

$$\begin{aligned} g(-\theta) &= \frac{\sin(-\theta)}{\cos(-\theta)} \\ &= \frac{-\sin(\theta)}{\cos(\theta)} \\ &= -\frac{\sin(\theta)}{\cos(\theta)} \end{aligned}$$

$$-g(\theta) = -\frac{\sin(\theta)}{\cos(\theta)}$$

$$g(-\theta) = -g(\theta) \implies g \text{ odd}$$

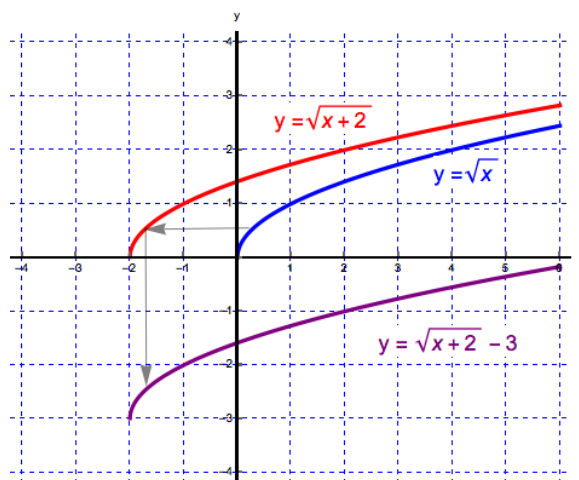
Problem 8

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Problem 8 Using the known graphs of $y = \sqrt{x}$ and $y = \frac{1}{x}$, sketch the graphs of the following using transformations.

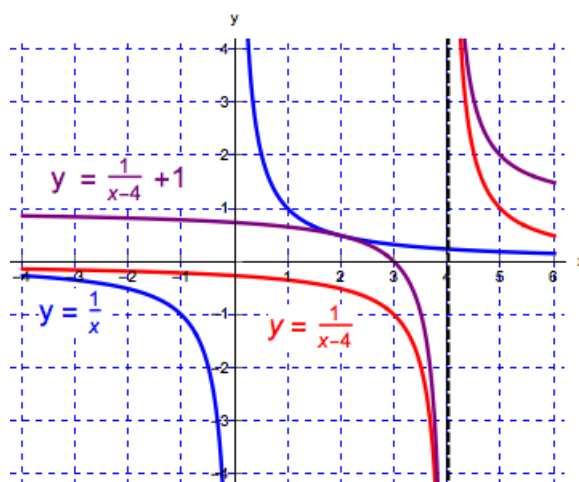
(a) $y = \sqrt{x+2} - 3$

Explanation. This is a shift of $y = \sqrt{x}$ moved left two units and down three units.



(b) $y = \frac{1}{x-4} + 1$

Explanation. This is a shift of $y = \frac{1}{x}$ moved right four units and up one unit.



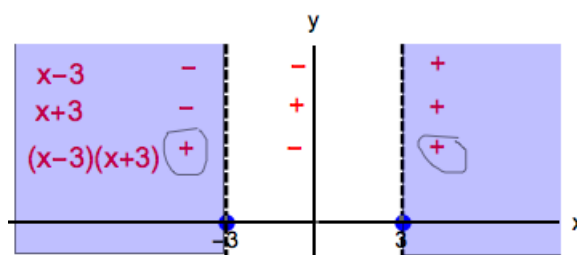
Problem 9

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Problem 9 Find the domain of the function. Determine whether the function is odd, even, or neither.

(a) $f(x) = \frac{x}{\sqrt{x^2 - 9}}$

Explanation. To find the domain, recall that a rational expression cannot have 0 in the denominator and a square root expression cannot have a negative number under the square root. Thus, $x^2 - 9 > 0 \implies (x - 3)(x + 3) > 0$. The zeros are located at $x = -3, 3$. From this we can draw a sign chart for the expression, $x^2 - 9$, and test values.



We see that $x^2 - 9 > 0$ on the interval $(-\infty, -3) \cup (3, \infty)$. Thus our domain is $(-\infty, -3) \cup (3, \infty)$.

Next we check for even/odd/neither.

$$f(-x) = \frac{-x}{\sqrt{(-x)^2 - 9}} \text{ which does not equal } f(x) \text{ so } f \text{ is not even}$$

$$-f(x) = -\frac{x}{\sqrt{x^2 - 9}} \text{ which is equal to } f(-x) \text{ so } f \text{ is odd.}$$

(b) $g(x) = \frac{\sin(x)}{x}$

Explanation. To find the domain, recall that a rational expression cannot have 0 in the denominator. Therefore, our domain is $(-\infty, 0) \cup (0, \infty)$

Next we check for even/odd/neither.

$$f(-x) = \frac{\sin(-x)}{-x} = \frac{-\sin(x)}{-x} = \frac{\sin(x)}{x} \text{ which equals } f(x) \text{ so } f \text{ is even}$$

(c) $h(t) = \ln(t^3 - 1)$

Explanation. To find the domain, recall that we cannot take the natural logarithm of 0 or a negative number. Therefore, $t^3 - 1 > 0 \implies (t - 1)(t^2 + t + 1) > 0$. The zero is located at $t = 1$. From this we can draw a sign chart and test values.



Problem 9

We see that $t^3 - 1 > 0$ on the interval $(1, \infty)$. Thus our domain is $(1, \infty)$.

Next we check for even/odd/neither.

$h(t)$ is neither even nor odd because if t is in the domain, then $-t$ is not in the domain.

Problem 10

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Problem 10

Problem 10 Let g be a one-to-one function and let g^{-1} be its inverse. **True or False:** If the point $(2, 1/5)$ lies on the graph of g , then the point $(2, 5)$ lies on the graph of g^{-1} .

Explanation. This statement is **false**: we have $g(2) = 1/5 \iff 2 = g^{-1}(1/5)$. The notation g^{-1} never, in this course, means $1/g$.

Problem 11

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Problem 11 Each of the following functions are invertible on their given domains. For each one find a formula for its inverse and give the domain and range of the inverse.

- (a) The function f defined by $f(x) = x^2 - 4x - 5$ for every $x \geq 2$.

Explanation. To help find the a formula for f^{-1} we will first “complete the square”:

$$\begin{aligned} x^2 - 4x - 5 &= x^2 - 4x + 4 - 4 - 5, \\ &= (x - 2)^2 - 9. \end{aligned}$$

Setting $y = f(x) = x^2 - 4x - 5 = (x - 2)^2 - 9$, we can follow the procedures outlined for algebraically finding the formula for an inverse function.

$$\begin{aligned} y &= (x - 2)^2 - 9 \\ \implies y + 9 &= (x - 2)^2 \\ \implies \sqrt{y + 9} &= |x - 2| \\ \implies \sqrt{y + 9} &= x - 2 \quad (\text{since } x \geq 2) \\ \implies \sqrt{y + 9} + 2 &= x \\ \implies 2 + \sqrt{x + 9} &= y \quad (\text{interchange } x \text{ and } y \text{ along with rewriting}) \end{aligned}$$

Therefore we have that f^{-1} is defined by $f^{-1}(x) = 2 + \sqrt{x + 9}$. The domain of $f^{-1}(x)$ is $[-9, \infty)$ and the range is $[2, \infty)$.

- (b) The function g defined by $g(u) = \sqrt[4]{u + 2}$.

Explanation. Following the procedure to algebraically find the formula for the inverse function we have

$$\begin{aligned} z &= \sqrt[4]{u + 2} \\ \implies z &= (u + 2)^{1/4} \\ \implies z^4 &= u + 2 \\ \implies z^4 - 2 &= u \\ \implies u^4 - 2 &= z \quad (\text{interchange } u \text{ and } z) \end{aligned}$$

Therefore we have that g^{-1} is defined by $g^{-1}(u) = u^4 - 2$. The domain of g^{-1} is $[0, \infty)$ and the range is $[-2, \infty)$.

- (c) The function h defined by $h(t) = 1/(t + 2)^2$ for every $t > -2$.

Explanation. Following the procedure to algebraically find the formula for the inverse function we have

$$\begin{aligned}s &= \frac{1}{(t+2)^2} \\ \implies (t+2)^2 &= \frac{1}{s} \\ \implies |t+2| &= \sqrt{\frac{1}{s}} \\ \implies t+2 &= \sqrt{\frac{1}{s}} \quad (\text{since } t > -2) \\ \implies t &= \sqrt{\frac{1}{s}} - 2 \\ \implies s &= \frac{1}{\sqrt{t}} - 2 \quad (\text{interchange } s \text{ and } t)\end{aligned}$$

Therefore we have h^{-1} is defined by $h^{-1} = \frac{1}{\sqrt{t}} - 2$. The domain of h^{-1} is $(0, \infty)$ and the range is $(-2, \infty)$.

Problem 12

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Problem 12 Explain what each of the following means:

(a) $f^{-1}(x)$

Explanation. This denotes the inverse function of f , f^{-1} , evaluated at x .

(b) $f(x^{-1})$

Explanation. This means $f\left(\frac{1}{x}\right)$.

(c) $(f(x))^{-1}$

Explanation. This means $f(x)$ raised to the -1 power, i.e. $\frac{1}{f(x)}$.

Problem 13

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Problem 13

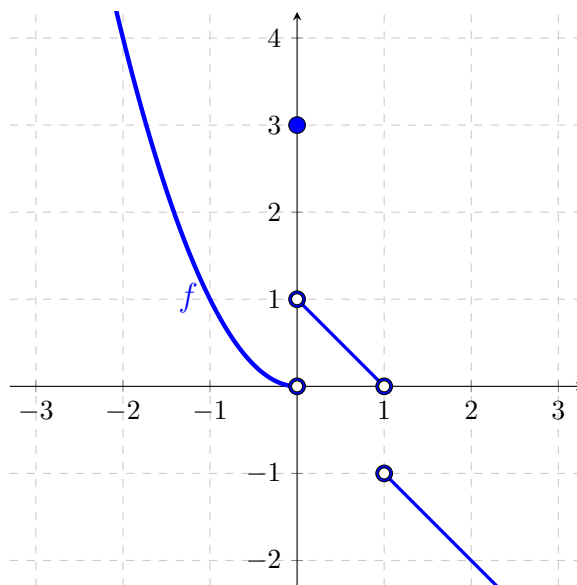
Problem 13 *If $f(x)$ represents the number of packages of buns needed for x packages of hotdogs, what does $f^{-1}(x)$ represent?*

Explanation. $f^{-1}(x)$ represents the numbers of packages of hotdogs needed for x packages of buns.

Problem 14

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Problem 14 We're given the following graph of a function:



Use this graph to answer the following questions:

- (a) What is the domain of this function?

Explanation. $(-\infty, 1) \cup (1, \infty)$

- (b) What is the range of this function?

Explanation. $(-\infty, -1) \cup (0, \infty)$

- (c) What is the value of $f(0)$, $f(1)$, and $f(2)$?

Explanation. $f(0) = 3$, $f(1)$ does not exist, $f(2) = -2$

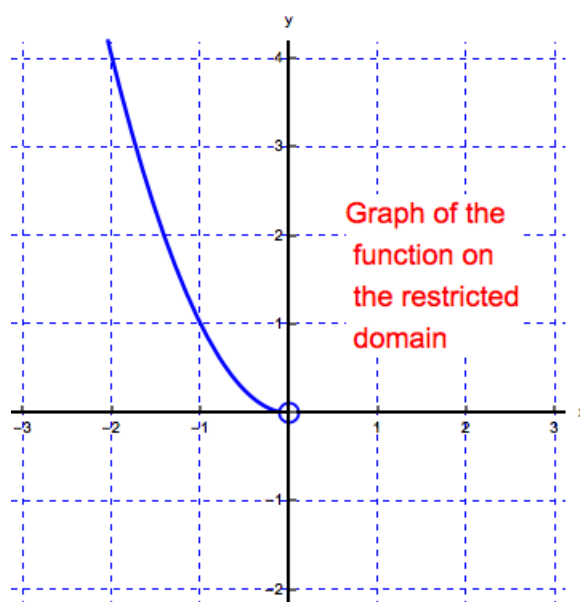
- (d) Does this function have an inverse? (Why or why not?)

Explanation. No, the function does not have an inverse. It is not one-to-one (that is, it does not pass the horizontal line test).

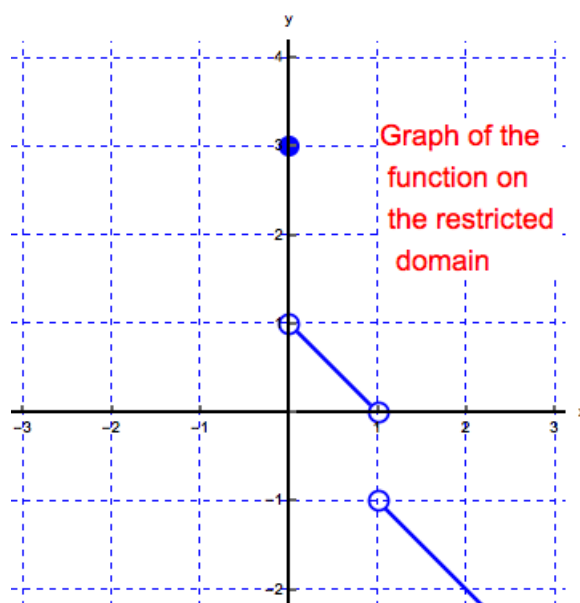
- (e) Find at least two intervals on which the function is one-to-one.

Explanation. The function becomes one-to-one when we restrict its domain to $(-\infty, 0)$:

Problem 14



The function also becomes one-to-one when we restrict its domain to $[0, 1) \cup (1, \infty)$:



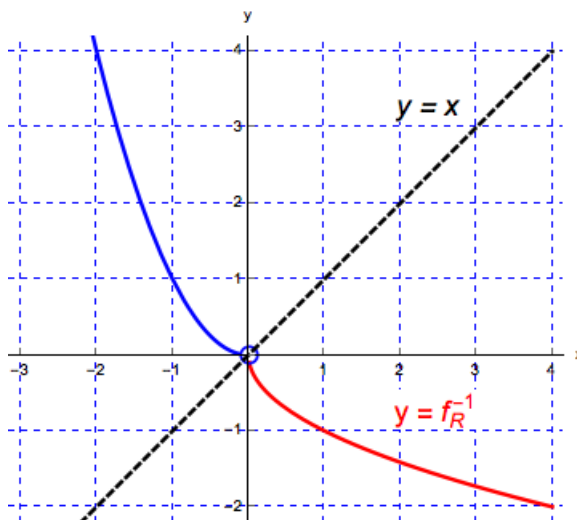
(f) Find $f^{-1}(3)$ on a restricted domain of f .

Explanation. In this case restrict the domain of f to $[0, 1) \cup (1, \infty)$. By definition we have $f^{-1}(3) = y \iff 3 = f(y)$. Looking at the second graph of the restricted function we see that $f(0) = 3$, that is, $f^{-1}(3) = 0$.

(If we restrict the domain of f to $(-\infty, 0)$, then $f^{-1}(3) = y \iff 3 = f(y)$ implies that $f(-1.7) \approx 3$. Hence $-1.7 \approx f^{-1}(3)$.)

- (g) Using the restricted domain for f of $(-\infty, 0)$, sketch a graph of f^{-1} .

Explanation. To graph the inverse, we can think of the graph being reflected over the line $y = x$. Another way to obtain the graph is to remember that if (x, y) is a point on the graph of f , then (y, x) is a point on the graph of f^{-1} . For example, $(-1, 1)$ is on the graph of f so $(1, -1)$ is on the graph of f^{-1} .



- (h) Using the restricted domain for f of $[0, 1) \cup (1, \infty)$, sketch a graph of f^{-1} .

Explanation. We can graph the inverse of f , when restricted to $[0, 1) \cup (1, \infty)$, in pieces from left to right. Since $(0, 3)$ is a point on the graph of f , the point $(3, 0)$ is on the graph of f^{-1} . Then, we see on the graph of f , there is a linear piece going from $(0, 1)$ to $(1, 0)$. When we imagine reflecting this part of the graph of f over the line $y = x$, it reflects onto itself. The last piece of the graph of f is a line from $(1, -1)$ which appears to also contain the point $(-2, 2)$. Reflecting this part of f over the line $y = x$, we obtain a line starting at $(-1, 1)$ and through the point $(-2, 2)$.

Problem 14

